

ISTA and FISTA - prox and acceleration in action

$$\min: f(x) = \underset{\substack{\uparrow \\ \text{smooth}}}{g(x)} + \underset{\substack{\uparrow \\ \text{"easy" prox operator}}}{h(x)} \quad \sim \text{convex}$$

subgradient method: $x_{t+1} = x_t - \eta g_t$, $g_t \in \partial f(x_t)$

Convergence: $O(1/\sqrt{\epsilon}) \sim O(1/\epsilon^2)$

proximal gradient: $x_{t+1} = \text{prox}_{\eta h}(x_t - \eta \nabla g(x_t))$

$\overset{\text{T}}{\text{ISTA}}$

Convergence: $O(1/t) \sim O(1/\epsilon)$

prox + acceleration (momentum) : Convergence $O(1/t^2) \sim O(1/\sqrt{\epsilon})$

FISTA

Lasso — L1 regularized regression

Vector $x \in \mathbb{R}^p$ p very large

We see x only through (noisy) linear measurements.

$$y_i = a_i^T x + \epsilon_i \quad : \quad \text{How many measurements do we need?}$$

Need: n measurements, $n > p$

Suppose: x has only s non-zeros — "sparsity"

↳ Compressive (compressed) sensing

(Under some technical assumptions: $n \sim s \log p / \epsilon^2$)

$$\hat{x} = \arg \min: \underbrace{\|Ax - y\|_2^2}_{\text{smooth}} + \underbrace{\lambda \|x\|_1}_{\text{convex but not smooth}} \quad \left(\|x\|_1 = \sum_{i=1}^p |x_i| \right) \quad \text{easy prox}$$

Sub-gradient descent

$$x_{t+1} = x_t - \eta (2A^T(Ax_t - y) + \lambda z) \rightarrow O(1/\sqrt{\epsilon})$$

$$z_i = \begin{cases} +1 & (x_t)_i > 0 \\ -1 & (x_t)_i < 0 \\ [-1, 1], & (x_t)_i = 0 \end{cases} \quad \begin{matrix} s \\ O(1/\epsilon^2) \end{matrix}$$

Proximal gradient - a simple interpretation

$$\min: f(x) = g(x) + h(x) = \|Ax - y\|_2^2 + \lambda \|x\|_1$$

$$\text{prox grad: } x_{t+1} = \text{prox}_{\eta h} (x_t - \eta \nabla g(x_t))$$

what should η be here?

$\eta = 1/\beta$, where g is β -smooth.

$$\begin{aligned} x_{t+1} &= \arg\min_x \lambda \|x\|_1 + \frac{\beta}{2} \|x - (x_t - \frac{1}{\beta} \nabla g(x_t))\|_2^2 \\ &= \arg\min_x (\lambda/\beta) \|x\|_1 + \frac{1}{2} \|x - (x_t - \frac{1}{\beta} \nabla g(x_t))\|_2^2 \\ &= \text{prox}_{(\lambda/\beta) \cdot \|\cdot\|_1} (x_t - \eta \nabla g(x_t)) \end{aligned}$$

Recall: $\text{prox}_{\alpha \cdot \|\cdot\|_1}(z) = u$, $u_i = (|z_i| - \alpha)_+ \cdot \text{sign}(z_i)$

Ex: $\alpha = 1.5$, $z = \begin{pmatrix} -2.5 \\ -0.7 \\ 2.2 \end{pmatrix}$, $\text{prox}_{\alpha \|\cdot\|_1}(z) = \begin{pmatrix} -1 \\ 0 \\ 0.7 \end{pmatrix}$

ISTA: $\text{prox}_{(\lambda/\beta) \|\cdot\|_1} (x_t - 2A^T(Ax - y)) \xrightarrow{\quad} O(1/t)$

FISTA - Fast ISTA (prox + acceleration)

$$\min: f(x) = g(x) + h(x)$$

$$\left(\mu_0 = 0, \quad \mu_t = \frac{1 + \sqrt{1 + 4\mu_{t-1}^2}}{2}, \quad \gamma_t = \frac{1 - \mu_t}{\mu_{t+1}} \right)$$

$$x_1 = z_1 = \text{arbitrary}$$

$$z_{t+1} = \text{prox}_{\frac{1}{\beta}h} \left(x_t - \frac{1}{\beta} \nabla g(x_t) \right)$$

$$x_{t+1} = (1 - \gamma_t) \cdot z_{t+1} + \gamma_t x_t$$

$$\text{For Lasso: } z_{t+1} = \text{prox}_{\lambda/\beta \|\cdot\|_1} \left(x_t - \frac{1}{\beta} \nabla g(x_t) \right)$$

$$2A^T(Ax_t - y).$$

$$(\text{conv rate: } O(1/t^2) \sim O(1/\sqrt{\epsilon})).$$